LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Vibhum Sharma
Roll Number	RSI2026506
Programme / Department	IT
Topic ID	T32
Full Assigned Topic	Assessing Misreading of Axis Scales and Units in AI-Based Chart Interpretation
AI Tool Used	Gemini 3.1 Pro
Date	21 August 2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	
Model name	
Model/version shown	
Access tier	☐ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☐ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☐ Yes ☐ No

Date/time generated	August 21, 2026 [09:40 AM]

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> *Generate a scholarly research paper on the topic: "*Assessing Misreading of Axis Scales and Units in AI-Based Chart Interpretation*". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Assessing Misreading of Axis Scales and Units in AI-Based Chart Interpretation

Measure	Value
Approximate pages	5
Approximate word count	1775
Number of sections	9
Total number of references	10
Approximate in-text citation occurrences	9 distinct in-text citations + Reference [10] is listed in the bibliography but has **no explicit in-text citation marker** anywhere in the body text.
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	0
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	5

URL only	0
No persistent identifier supplied	5
TOTAL	10

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	[1] Masry et al. (2022) — ChartQA
R2	[2] Liu et al. (2023) — DePlot
R3	[3] Han et al. (2023) — ChartLlama
R4	[4] Zhang et al. (2024) — CharXiv
R5	[5] Chen et al. (2024)
R6	[6] Li et al. (2025) — ChartSketcher
R7	[7] Wu et al. (2025) — ChartInsighter
R8	[8] Lim et al. (2025) — ChartCap
R9	[9] Kim et al. (2026)
R10	[10] Anonymous (2026) — ChartAttack

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine

5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Well-known venue (ACL Findings), specific page range, clean metadata.	5	Yes
R2	Precise versioned arXiv ID, plausible author/topic pairing.	5	Yes
R3	Specific versioned arXiv ID, topic fits naturally into the field.	5	Yes
R4	Vague "Preprint / ResearchGate" sourcing, no real identifier.	2	No
R5	Specific arXiv ID, consistent phrasing/topic fit.	4	Yes
R6	Real, well-known venue (NeurIPS), but no arXiv/DOI given.	3	Yes
R7	Legitimate-sounding journal (IEEE TVCG), no identifier given.	3	Yes
R8	Specific versioned arXiv ID, consistent naming/topic fit.	4	Yes
R9	"OpenReview" alone, no ID/link, unusually recent (2026) date.	2	No
R10	"Anonymous" author, dual/ambiguous source tag, forward-dated-looking ID, never cited in body text.	1	No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	NA
R2	Y	Y	N	Y	Y	Y
R3	Y	Y	N	Y	Y	Y
R4	Y	Y	N	Y	Y	NA
R5	Y	Y	N	Y	Y	Y
R6	Y	Y	N	Y	Y	NA
R7	Y	N	N	Y	Y	NA
R8	Y	Y	Y	Y	Y	Y
R9	N	N	N	N	N	NA
R10	Y	Y	N	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	ACL Anthology, ACL Anthology BibTeX, GitHub (vis-nlp/ChartQA)	DOI 10.18653/v1/2022.findings-acl.177	Exact match on title, authors (Masry, Long, Tan, Joty, Hoque), venue, and pages 2263–2279.
R2	arXiv abstract page, Semantic Scholar	arXiv:2212.10505v2	Real paper, correct title/ID/year. But cited authors "Masi, F. D." and "Das, A." don't exist — real co-authors include Piccinno, Krichene, Pang, Lee, Joshi, Chen, Collier, Altun instead.
R3	arXiv abstract page, Semantic Scholar	arXiv:2311.16483	Real paper, correct title/ID/year/venue. Real first author is Yucheng **Han** (not

			"Han, J.") and third author is Xin **Chen** (not "Chen, J.").
R4	OpenReview (NeurIPS 2024), Princeton Collaborate, official GitHub	No ID given by AI (only "Preprint/ResearchGate")	Real paper — title matches exactly, and even the specific stats cited in the body (GPT-4o 47.1%, human 80.5%) are correct. But real first author is Zirui **Wang**, not "Zhang, M." — completely wrong author. Real arXiv ID is 2406.18521, which the AI never supplied.
R5	arXiv abstract page, IEEE Xplore, dblp	arXiv:2407.17291	Real paper, correct title/ID/year. Real authors are Leo Yu-Ho **Lo** and Huamin **Qu** — not "Chen, Y." at all.
R6	OpenReview (NeurIPS 2025), dblp, official GitHub	No ID given by AI (real one is arXiv:2505.19076)	Real paper, correct title/venue/year. Real first author is Muye **Huang**, not "Li, L."
R7	arXiv abstract page, IEEE Xplore, official GitHub	No ID given by AI (real one is arXiv:2501.09349)	Real paper exists, but AI's title is truncated (drops "Generation with A Benchmark Dataset"). Real authors are Fen **Wang** et al. — not "Wu, H."
R8	arXiv abstract page, Hugging Face, ResearchGate	arXiv:2508.03164	Exact match: title, all three authors (Lim, Ahn, Kim), year, identifier.
R9	Multiple targeted web searches (title phrase, "87.1%" statistic, "explanation-level hallucination" + chart)	None findable	No trace of this paper anywhere — not on OpenReview, arXiv, Semantic Scholar, or Google Scholar-indexed sources. The specific "87.1%" statistic doesn't appear in any real paper either.
R10	arXiv abstract page, official GitHub (insait-institute/chartAttack)	arXiv:2601.12983	Real paper, correct ID, close title match (real title ends "...in Chart Generation," not "/ChartAttack"). **Not anonymous** — real named authors are Ortiz-Barajas, Tonglet, Gupta, and Gurevych. Also worth noting: submitted Jan 2026, so it's *not* forward-dated as I'd guessed in Part F.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	
R2	B	
R3	B	
R4	B	
R5	B	
R6	B	
R7	B	
R8	A	
R9	D	
R10	B	

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 2 B = 7 C = 0 D = 1 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(2) + 3(7) + 1(0) + 0(1) + 1(0) = 29 points

Authenticity Score = 29 / 40 x 100 = 72.5 /100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 8 Total predictions = 10 Prediction Accuracy = 80 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	10
References deeply audited	10
A: Verified	2
B: Wrong metadata	7
C: Frankenstein	0
D: Fabricated	1
E: Identifier mismatch	0
Authenticity Score	72.5 /100
Prediction Accuracy	80 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Most of the citations were not reliable. Every reference with a clean, versioned arXiv ID pointed to a real paper but 4 of those still contained fabricated or incorrect metadata.

2. Did all genuine references actually support the claims for which they were cited?

I audited citation authenticity exhaustively across all citations, but most of it were not supporting the claims for which they are cited.

3. What was the most serious citation-integrity problem you observed?

Systematic author fabrication, not missing papers, but wrong people attached to real ones. 7 of the 10 references (70%) had incorrect authors: invented names that don't exist.

4. What is the single most important lesson you learned from this lab?

Verifying that a citation exists is necessary but not sufficient you also have to verify who wrote it.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Vibhum Sharma

Roll Number: RSI2026506

Signature: Vibhum Sharma

Date: 21 August, 2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission

Assessing Misreading of Axis Scales and Units in AI-Based Chart Interpretation

Abstract
As Multimodal Large Language Models (MLLMs) and Vision-Language Models (VLMs) become increasingly integrated into automated data analysis, their ability to accurately interpret data visualizations is paramount. Despite achieving high accuracy on standardized benchmarks, these models frequently struggle with the precise spatial grounding and numerical extraction required to read complex charts. This paper investigates a critical point of failure in AI-based chart interpretation: the systematic misreading of axis scales and units. By examining current literature on multimodal chart comprehension, this study highlights how features such as logarithmic scales, dual axes, non-standard units, and dense tick marks trigger explanation-level hallucinations and data misalignment. We review current interventions—such as plot-to-table translation modules and sketch-based reasoning frameworks—and outline the persistent research gaps in evaluating explanation faithfulness over mere answer correctness.
Keywords: Multimodal Large Language Models (MLLMs), Chart Understanding, Vision-Language Models (VLMs), Hallucination, Axis Misinterpretation, Data Visualization

1. Introduction

Charts and plots are high-density visualization carriers for complex data, serving as crucial mediums for information extraction in scientific, financial, and business contexts. The advent of Multimodal Large Language Models (MLLMs), such as GPT-4o, Gemini, and open-weight equivalents like InternVL and Qwen-VL, has catalyzed a shift toward automated chart interpretation. These systems can answer queries, summarize trends, and extract data directly from bitmap images of charts.

However, extracting semantic meaning from charts necessitates more than natural language processing; it requires precise visual-spatial reasoning. A persistent vulnerability in state-of-the-art MLLMs is their inability to accurately read and map axis scales, tick marks, and units to the corresponding data series. Even when models arrive at the correct final answer, they frequently fabricate the intermediate steps of reading the axes—a phenomenon known as explanation-level hallucination. This paper provides a scholarly assessment of how and why AI models misread axis scales and units, evaluates current technical mitigations, and identifies critical limitations in how chart understanding is currently benchmarked.

2. Background and Literature Review

The evaluation of chart understanding in AI initially relied on template-based, procedurally generated datasets with limited visual diversity. Early benchmarks measured basic perceptual tasks, such as identifying chart types or retrieving explicit values.

The introduction of **ChartQA** by Masry et al. (2022) marked a paradigm shift by incorporating human-authored, complex reasoning questions that require both visual and logical synthesis. ChartQA revealed that while models could identify general trends, they struggled significantly with tasks requiring arithmetic operations based on visual estimations from axes.

Recent literature underscores that standard evaluation suites often overestimate model capabilities. Zhang et al. (2024) introduced **CharXiv**, a benchmark derived from real-world scientific papers, revealing a substantial, previously underestimated gap between proprietary models (e.g., GPT-4o achieving 47.1% accuracy) and human performance (80.5%) on reasoning tasks. A primary driver of this performance gap is the failure of models to reliably interpret structural elements like legends, units, and axes. Furthermore, Chen et al. (2024) demonstrated in their assessment of misleading visualizations that LLMs are vulnerable to misinterpreting spatial representations when axes are manipulated.

3. Main Thematic and Technical Discussion

The misreading of axes by MLLMs is not a monolithic error but manifests through several specific visual-reasoning failures.

3.1. The Vulnerability of Logarithmic Scales

Standard linear axes are generally well-comprehended by top-tier MLLMs because the spatial distance between tick marks correlates directly and proportionally with numerical values. However, logarithmic scales heavily disrupt the heuristic spatial assumptions of VLMs. When interpreting log-scaled axes, MLLMs frequently map data points using linear interpolation, leading to massive numerical extraction errors. This stems from the models' over-reliance on learned geometric patterns rather than reading the explicit mathematical relationship defined by the tick mark labels.

3.2. Dual Axis Interpretability and Data Misalignment

Dual-axis charts (charts containing a primary y-axis on the left and a secondary y-axis on the right) present a severe binding problem for VLMs. Models must not only read the axes but associate specific data series (e.g., a bar chart and a line plot superimposed) with the correct axis. Studies testing the vulnerability of LLMs to chart manipulation indicate that dual-axis configurations frequently lead to misinterpretation, as the model incorrectly maps the magnitude of one data series to the scale of the opposing axis. This failure in visual-semantic grounding results in models falsely claiming correlations or magnitude differences that do not exist in the underlying data table.

3.3. Unit Conversion and Tick-Mark Hallucinations

A frequent limitation in AI chart interpretation is the precise reading of dense numerical tick marks. When data points lie between explicit tick marks, humans use proportional spatial estimation. VLMs, however, often struggle to interpolate these values accurately. Lim et al. (2025) noted in their work on the ChartCap framework that "axis hallucinations" account for a significant portion of errors in dense chart captioning. Furthermore, units of measurement (e.g., millions, percentages, specific scientific metrics) are often presented in abbreviated forms on the axis label. If the model fails to optically extract or semantically prioritize the axis title, it defaults to raw numerical outputs, resulting in a scale magnitude error (e.g., reporting "5" instead of "5 million").

4. Current Approaches and Developments

To circumvent the visual-spatial limitations of purely end-to-end MLLMs, the field has seen several innovative developments aimed at bridging the gap between pixel space and numerical reasoning.

4.1. Modality Conversion: Plot-to-Table Translation

One of the most effective interventions is bypassing direct visual reasoning through modality conversion. Liu et al. (2023) introduced **DePlot**, a one-shot visual language reasoning module that translates the image of a plot into a linearized data table. By first derendering the chart into structural text, the system relies on the robust numerical and logical reasoning capabilities of standard LLMs to answer queries, effectively bypassing the visual axis-reading bottleneck.

4.2. Visual Grounding and Sketch-Based Reasoning

Other approaches attempt to fix the visual reasoning process itself. **ChartLlama** (Han et al., 2023) utilized specialized instruction-tuning datasets to improve the model's fundamental grounding of chart elements.

More recently, Li et al. (2025) proposed **ChartSketcher**, which addresses the limitation of textual Chain-of-Thought (CoT) by enabling MLLMs to explicitly annotate intermediate reasoning steps directly onto the chart image. By drawing on the image (e.g., projecting a line from a data point to the y-axis to read the scale), the model leverages visual feedback and reflection, mimicking the human subconscious strategy of tracing lines to axes for accurate value estimation.

4.3. Multi-Agent and Verification Frameworks

To address temporal and trend hallucinations, frameworks like **ChartInsighter** (Wu et al., 2025) utilize multi-agent iterative collaboration and self-consistency testing. By combining visualization specifications with the underlying data tables, these systems cross-verify the LLM's generated summary against the raw data, explicitly mitigating extremum errors and trend direction hallucinations.

5. Research Challenges and Limitations

Despite advancements, evaluating and mitigating axis and scale misinterpretations faces profound structural challenges within the research community.

5.1. Explanation-Level Hallucination

As VLMs increasingly produce free-form explanations for chart reasoning, researchers have discovered a critical flaw: models can provide the correct final answer while entirely fabricating the axis-reading steps. Kim et al. (2026) defined this as "explanation-level hallucination". In a study of ChartQA instances, they found that 87.1% of QA-correct responses from base VLMs contained claims unsupported by visual evidence—meaning the model guessed or inferred the correct answer without faithfully reading the axis or tick marks.

5.2. Benchmark Blindness

This high rate of hallucination exposes a limitation known as "Benchmark Blindness". Standard evaluation suites grade models primarily on answer extraction accuracy (often using a relaxed numerical threshold). Because the evaluation function projects only the final token and discards the reasoning, benchmark scores are inflated, masking the model's fundamental inability to read scales and units reliably.

6. Research Gaps and Future Directions

The current trajectory of chart interpretation research suggests several critical gaps that must be addressed:

1. **Faithfulness as an Independent Metric:** Future benchmarks must adopt claim-level verification to ensure that intermediate visual reasoning (e.g., reading an axis intercept) is as faithful as the final answer.
2. **Specialized Datasets for Non-Linear Scales:** Existing training corpora are heavily biased toward simple, linear bar and line charts. There is a pressing need for curated datasets focusing explicitly on dual axes, logarithmic scales, and broken axes to enforce robust spatial grounding.
3. **Integration of Metadata and Source Verification:** As seen in studies of biological and scientific charts, LLMs often hallucinate conclusions based on their training priors rather than the visual evidence. Future systems must better separate the extraction of visual facts (what the axis actually says) from the model's semantic priors (what the model expects the axis to say).

7. Conclusion

The automated interpretation of charts by Multimodal Large Language Models holds transformative potential for data analysis. However, as this paper demonstrates, the precise extraction of numerical data via axis scales and units remains a fragile point in the AI reasoning pipeline. MLLMs consistently struggle with logarithmic scales, dual-axis misalignment, and tick-mark interpolation, often hiding these perceptual failures behind explanation-level hallucinations. While innovations like plot-to-table translation (DePlot) and multimodal feedback (ChartSketcher) show great promise in mitigating these errors, the community must evolve its evaluation metrics. Until benchmark blindness is resolved and explanation faithfulness is explicitly optimized, the vulnerability of AI models to misreading axis scales will remain a significant barrier to their autonomous deployment in critical analytical settings.

Complete References

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